4. MULTI VARIABLE CALCULUS [PARTIAL DIFFERENTIATION]

Limits & continuty:

Q) Evaluate $\lim\limits_{\substack{x\to 1\\ y\to 2}} \frac{2x^2y}{x^2+y^2+1}$

Sol) $\lim\limits_{y\to 2}\left\{\lim\limits_{x\to 1} \frac{2x^2y}{x^2+y^2+1}\right\} = \lim\limits_{y\to 2}\left\{\frac{2y}{y^2+2}\right\} = \frac{4}{6} = \frac{2}{3}$

$\lim\limits_{x\to 1}\left\{\lim\limits_{y\to 2} \frac{2x^2y}{x^2+y^2+1}\right\} = \lim\limits_{x\to 1}\left\{\frac{4x^2}{x^2+5}\right\} = \frac{4}{6} = \frac{2}{3}$

It exists.

Q) $\lim\limits_{\substack{x\to 0\\ y\to 0}} \frac{x-y}{2x+y}$

Sol) $\lim\limits_{y\to 0}\left\{\lim\limits_{x\to 0} \frac{x-y}{2x+y}\right\} = \lim\limits_{y\to 0}\left\{\frac{-y}{y}\right\} = \lim\limits_{y\to 0} -1 = -1$

$\lim\limits_{x\to 0}\left\{\lim\limits_{y\to 0} \frac{x-y}{2x+y}\right\} = \lim\limits_{x\to 0}\left\{\frac{x}{2x}\right\} = \frac{1}{2}$

Q) Discuss the continuty of function $f(x,y) = \frac{2xy}{x^2+y^2}$ $(x,y) \neq (0,0)$; 0; $(x,y) = (0,0)$

Sol) $f(x) = \begin{cases} \frac{2xy}{x^2+y^2} & ; (x,y) \neq (0,0) \\ 0 & ; (x,y) = (0,0) \end{cases}$

$f(a,b) = f(0,0) = 0$

1) $\lim\limits_{x\to 0}\left\{\lim\limits_{y\to 0} \frac{2xy}{x^2+y^2}\right\} = \lim\limits_{x\to 0} \frac{0}{x^2} = 0 = f(0,0)$

2) $\lim\limits_{y\to 0}\left\{\lim\limits_{x\to 0} \frac{2xy}{x^2+y^2}\right\} = \lim\limits_{y\to 0} \frac{0}{y^2} = 0 = f(0,0)$

3) along $y = mx$

$\lim\limits_{x\to 0} \frac{2x(mx)}{x^2+m^2x^2} = \lim\limits_{x\to 0} \frac{2x^2m}{x^2(1+m^2)}$ $\lim\limits_{x\to 0} \frac{2m}{1+m^2}$

$\frac{2m}{1+m^2} \neq f(0,0)$

Q) Discuss the continuity of $f(x,y) = \begin{cases} \frac{x^2}{\sqrt{x^2+y^2}} & ; (x,y) \neq (0,0) \\ & ; (x,y) = (0,0) \end{cases}$

Sol) $f(a,b) = f(0,0)$

① $\lim_{x\to 0}\left\{\lim_{y\to 0} \frac{x^2}{\sqrt{x^2+y^2}}\right\} = \lim_{x\to 0} \frac{x^2}{x} = x$

② $\lim_{y\to 0}\left\{\lim_{x\to 0} \frac{x^2}{\sqrt{x^2+y^2}}\right\} = \lim_{y\to 0} \frac{0}{\sqrt{0+y^2}} = 0 = f(0,0)$

③ along $y=mx$

$\lim_{x\to 0} \frac{x^2}{\sqrt{x^2+m^2x^2}} = \lim_{x\to 0} \frac{x^2}{x\sqrt{1+m^2}} = \lim_{x\to 0} \frac{x}{\sqrt{1+m^2}} = 0 = f(0,0)$

④ along $y=mx^n$

$\lim_{x\to 0} \frac{x^2}{\sqrt{x^2+m^2x^{2n}}} = \lim_{x\to 0} \frac{x^2}{x\sqrt{1+m^2x^{2n-2}}} = \lim_{x\to 0} \frac{x}{\sqrt{1+m^2x^{2n-2}}} = 0$

Partial differentiation:

Differential equation

ordinary differential equation(d) → function of single variable → Dependent variable, Independent variable

partial differential equation(∂) → function of 2 (or) more varia → 1 dependent variable, 2 or more I d V

$T(t)$ — T → dependent, t → independents

$\frac{d(T)}{dt}$

$A(l,b)$

$\frac{\partial A}{\partial l}, \frac{\partial A}{\partial b} = \frac{\partial^2 A}{\partial l \partial b}$

$V(x,y,z)$

→ Dependent variable derivate ≠ 0

→ The derivative of 1 independent variable w.r.t another independent variable = 0

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$$\left\{\begin{array}{l} d(a\cdot x^n) = a\cdot nx^{n-1} \\ d(a+x^n) = 0+nx^{n-1} \end{array}\right\}$$

Q) Prove that $\frac{\partial^2 f}{\partial x^2 y} = \frac{\partial^2 f}{\partial y^2 dx}$ $\quad ax^2+by^2+2hxy$

Sol) let $f = ax^2+by^2+2hxy$

$$\frac{\partial f}{\partial x} = 2ax+2hy$$

$$\frac{\partial f}{\partial y} = 2by+2hx$$

$$\frac{\partial^2 f}{\partial x\cdot\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(2hy + 2hx)$$
$$= 0+2h(1) = 2h$$

$$\frac{\partial^2 f}{\partial y\cdot\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}(2ax+2hy) = 0+2h(1) = 2h$$

LHS = RHS

Q) Prove that $\frac{\partial^2 f}{\partial x\cdot\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}$; $f = x^3+y^3-3axy$

Sol) let $f = x^3+y^3-3axy$

$$\frac{\partial f}{\partial x} = 3x^2-3ay = 3(x^2-ay)$$

$$\frac{\partial f}{\partial y} = 0+3y^2-3ax = 3(y^2-ax)$$

LHS: $\frac{\partial^2 f}{\partial x\cdot\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(3(y^2-ax)\right)$

RHS: $\frac{\partial^2 f}{\partial y\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}\left(3(x^2-ay)\right)$

Q) If $\omega = (y-z)(z-x)(x-y)$ find $\frac{\partial \omega}{\partial x} + \frac{\partial \omega}{\partial y} + \frac{\partial \omega}{\partial z}$

Sol) $\frac{\partial \omega}{\partial x} = (y-z)\frac{\partial}{\partial x}\left((z-x)(x-y)\right)$

$= (y-z)\frac{\partial}{\partial x}\{zx - zy - x^2 + xy\}$

$= (y-z)\{z - 2x + y\}$

$= yz - 2xy + y^2 - z^2 + 2zx - zy = -2xy + y^2 - z^2 + 2zx$

$\frac{\partial \omega}{\partial y} = (z-x)\frac{\partial}{\partial y}\{(y-z)(x-y)\} = (z-x)\frac{\partial}{\partial y}\{xy - y^2 - zx + zy\}$

$= (z-x)\{x - 2y + z\} = zx - 2zy + z^2 - x^2 + 2xy - xz$

$= -2zy - x^2 + z^2 + 2xy$

$\frac{\partial \omega}{\partial z} = (x-y)\frac{\partial}{\partial z}\{(y-z)(z-x)\} = (x-y)\frac{\partial}{\partial z}\{yz - xy - z^2 + zx\}$

$= (x-y)\{y - 2z + x\} = xy - 2zx + x^2 - y^2 + 2zy - xy$

$= -2zx + x^2 - y^2 + 2zy$

$\frac{\partial \omega}{\partial x} + \frac{\partial \omega}{\partial y} + \frac{\partial \omega}{\partial z} = -2xy + y^2 - z^2 + 2zx - 2zy - x^2 + z^2 + 2xy - 2zx + x^2 - y^2 + 2zy = 0$

Q) If $U = \frac{1}{\sqrt{x^2+y^2+z^2}}$ prove that $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 0$ where $x^2+y^2+z^2 \neq 0$

Sol) $U = \frac{1}{\sqrt{x^2+y^2+z^2}} \quad (x^2+y^2+z^2)$

$\frac{\partial U}{\partial x} = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2}(2x+0+0) = -x(x^2+y^2+z^2)^{-3/2}$

$\frac{\partial^2 U}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial U}{\partial x}\right) = \frac{\partial}{\partial x}\left[-x(x^2+y^2+z^2)^{-3/2}\right]$

$\left[-x\left(-\frac{3}{2}(x^2+y^2+z^2)^{-5/2}\cdot 2x\right) - (x^2+y^2+z^2)^{-3/2}\cdot 1\right\}$

$3x^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$

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$$-(x^2+y^2+z^2)^{-3/2}$$

similarly $\frac{\partial^2 U}{\partial y^2} = 3y^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$

$$\frac{\partial^2 U}{\partial z^2} = 3z^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 0$$

$$3(x^2+y^2+z^2)^{-5/2}\ (x^2+y^2+z^2) - 3(x^2+y^2+z^2)^{-3/2}$$

$$3(x^2+y^2+z^2)^{-3/2} - 3(x^2+y^2+z^2)^{-3/2} = 0$$

Q) If $U = \log(x^3+y^3+z^3-3xyz)$ prove that $\left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right]U^2 = \frac{-9}{(x+y+z)^2}$

sol) $\frac{\partial U}{\partial x} = \frac{1}{(x^3+y^3+z^3-3xyz)} \times [3x^2+0+0-3yz]$

$$\frac{3(x^2-yz)}{x^3+y^3+z^3-3xyz}$$

similarly $\frac{\partial U}{\partial y} = \frac{3(y^2-zx)}{x^3+y^3+z^3-3xyz}$

consider $\left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right]^2 U$

$$\left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right]\left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right]U$$

$$\left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right]\left[\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} + \frac{\partial U}{\partial z}\right]$$

$$\left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right]\frac{3(x^2+y^2+z^2-xy-zy-zx)}{x^3+y^3+z^3-3xyz}$$

$$\left[\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right]\frac{3}{x+y+z}$$

$$\frac{\partial}{\partial x}\left(\frac{3}{x+y+z}\right) + \frac{\partial}{\partial y}\left(\frac{3}{x+y+z}\right) + \frac{\partial}{\partial z}\left(\frac{3}{x+y+z}\right)$$

$$\frac{-3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} = \frac{-9}{(x+y+z)^2}$$

Q) If $x^x+y^y+z^z=c$ then prove that at $x=y=z$ $\frac{\partial^2 z}{\partial x \partial y} = -[x\log ex]^{-1}$

Sol) $x^x+y^y+z^z=c$; $\frac{\partial^2 z}{\partial x \partial y} = -[x \log ex]^{-1}$

$x\log x + y\log y + z\log z = 1$

differentiating partially w.r.t.x

$[x \cdot \frac{1}{x} \log x \cdot 1] + 0 + [z \cdot \frac{1}{z} \frac{\partial z}{\partial x} + \log z \frac{\partial z}{\partial x}] = 0$

$(1+\log x) + \frac{\partial z}{\partial x}(1+\log z) = 0$

$\frac{\partial z}{\partial x} = \frac{1+\log x}{1+\log z}$

similarly $\frac{\partial z}{\partial y} = \frac{-(1+\log y)}{1+\log z}$

consider $\frac{\partial^2 z}{\partial x \cdot \partial y}$

$\frac{\partial}{\partial x}\left[\frac{\partial z}{\partial y}\right] = \frac{\partial}{\partial x}\left[\frac{-(1+\log y)}{1+\log z}\right] = -(1+\log y)\frac{\partial}{\partial x}\left[\frac{1}{1+\log z}\right]$

$= -(1+\log y)\left[-\frac{1}{(1+\log z)^2}\left(0+\frac{1}{z}\frac{\partial z}{\partial x}\right)\right] = -\frac{1}{z}\frac{(1+\log y)(1+\log x)}{(1+\log z)^2}$

at $x=y=z$ = $\frac{\partial^2 z}{\partial x \cdot \partial y} = -\frac{1}{x}\frac{(1+\log x)^2}{(1+\log x)^3}$

$= -\frac{1}{x}\frac{1}{1+\log x}$

$= \frac{-1}{x}\frac{1}{\log e + \log x}$

$= \frac{-1}{x\log ex}$

$= -[x\log ex]^{-1}$

Independent

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HOMOGENEOUS FUNCTIONS

A function x, y is said to be homogeneous of order n if $f(kx, ky) = k^n f(x, y)$

eg: $f(x,y) = x^2+y^2+2xy$

$f(kx, ky) = k^2x^2+k^2y^2+2kx\,2ky = k^2[x^2+y^2+2xy]$

homogeneous of degree $= k^2 f(x,y)$

Euler's theorem

If $z = f(x,y)$ is a homogeneous function of degree n then

$$x\cdot\frac{\partial z}{\partial x} + y\cdot\frac{\partial z}{\partial y} = nz$$

Q) If $u = \frac{x^3y^3}{x^3+y^3}$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$

Sol) $x(kx, ky) = \frac{k^3x^3\cdot k^3y^3}{k^3x^3+k^3y^3} = \frac{k^6(x^3y^3)}{k^3(x^3+y^3)} = k^3 u(x,y)$

$u(x,y)$ is a homogeneous function of degree 3

$$u = \frac{x^3y^3}{x^3+y^3}$$

differentiating u partially w.r.t x

$$\frac{du}{dx} = \frac{(x^3+y^3)\cdot y^3(3x^2) - x^3y^3(3x^2+0)}{(x^3+y^3)^2}$$

$$= \frac{3x^2y^3\,[x^3+y^3-x^3]}{(x^3+y^3)^2}$$

similarly $\frac{\partial u}{\partial y} = \frac{3x^6y^2}{(x^3+y^3)^2}$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$$

$$\frac{3x^3y^6}{(x^3+y^3)^2} + \frac{3x^6y^3}{(x^3+y^3)^2}$$

$$\frac{3x^3y^3}{(x^3+y^3)^2}(y^3+x^3) = 3\frac{x^3+y^3}{x^3+y^3} = 3u$$

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$xy+yz+zx$

Q) verify Euler's theorem for

Sol) let $f = xy+yz+zx$

$f(kx,ky) = kx\cdot ky + ky\cdot kz + kz\cdot kx = k^2[xy+yz+zx] = k^2 f(x,y)$

f is homogeneous function of degree $\boxed{n=2}$

differentiating f partially w.r.t. x

$\frac{\partial f}{\partial x} = y+z$; $\frac{\partial f}{\partial y} = x+z$; $\frac{\partial f}{\partial z} = y+x$

$x\frac{\partial f}{\partial x} + y\cdot\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = x(y+z)+y(x+z)+z(y+x)$

$= 2(xy+yz+zx)$

$= 2f$

Hence proved

Q) verify Euler's theorem for $u = x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)$

Sol) let $u(kx,ky) = k^2x^2\tan^{-1}\left(\frac{ky}{kx}\right) - k^2y^2\tan^{-1}\left(\frac{kx}{ky}\right)$

$= k^2\left[x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)\right]$

$= k^2 U = k^2 U(x,y)$

$U(x,y)$ is homogeneous function of degree $\boxed{n=2}$

differentiating u partially w.r.t. x

$\frac{\partial U}{\partial x} = \left[x^2\frac{1}{1+\frac{y^2}{x^2}}\left(-\frac{y}{x^2}\right) + \tan^{-1}\left(\frac{y}{x}\right)\cdot 2x\right] - y^2\left[\frac{1}{1+\frac{x^2}{y^2}}\cdot\frac{1}{y}\right]$

$= \left[x^2\frac{x^2}{x^2+y^2}\cdot\frac{-y}{x^2} + \tan^{-1}\left(\frac{y}{x}\right)\cdot 2x\right] - y^2\left[\frac{y^2}{y^2+x^2}\cdot\frac{1}{y}\right]$

$= \left[-\frac{x^2y}{x^2+y^2} + 2x\cdot\tan^{-1}\left(\frac{y}{x}\right) - \frac{y^3}{x^2+y^2}\right]$

$= \left[\frac{-x^2y-y^3}{x^2+y^2} + 2x\tan^{-1}\left(\frac{y}{x}\right)\right]$

$$= -y\frac{(x^2+y^2)}{x^2+y^2} + 2x\tan^{-1}\left(\frac{y}{x}\right)$$

$$= -y + 2x\tan^{-1}\left(\frac{y}{x}\right)$$

$$x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)$$

$$\frac{\partial U}{\partial y} = \left[x^2 \frac{1}{1+\frac{y^2}{x^2}}\cdot\frac{1}{x}\right] - \left[y^2\cdot\frac{1}{1+\frac{x^2}{y^2}}\left(-\frac{x}{y^2}\right) + \tan^{-1}\left(\frac{x}{y}\right)\cdot 2y\right]$$

$$= \left[x^2\cdot\frac{x^2}{x^2+y^2}\cdot\frac{1}{x}\right] - \left[y^2\ \frac{y^2}{x^2+y^2}\cdot -\frac{x}{y^2} + 2y\tan^{-1}\left(\frac{x}{y}\right)\right]$$

$$= \left[\frac{x^3}{x^2+y^2}\right] - \left[\frac{-xy^2}{x^2+y^2} + 2y\tan^{-1}\left(\frac{x}{y}\right)\right]$$

$$= \left[\frac{x(x^2+y^2)}{x^2+y^2} - 2y\tan^{-1}\left(\frac{x}{y}\right)\right]$$

$$= \left[x - 2y\tan^{-1}\left(\frac{x}{y}\right)\right]$$

$$x\frac{\partial U}{\partial x} + y\frac{\partial U}{\partial y}$$

$$= -xy + 2x^2\tan^{-1}\left(\frac{y}{x}\right) + xy - 2y^2\tan^{-1}\left(\frac{x}{y}\right)$$

$$= 2\left[x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)\right]$$

$$= 2U$$

Hence proved

Q) Q) $u = \log \frac{x^2+y^2}{x+y}$ P.T $x \cdot \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$

sol' sol)

$$e^u = \frac{x^2+y^2}{x+y}$$

let $z = e^u = \frac{x^2+y^2}{x+y}$

clearly z is a homogeneous function of degree $n = 1$

$\therefore$ By Euler's theorem

$$x \cdot \frac{\partial z}{\partial x} + y \cdot \frac{\partial z}{\partial y} = nz = z \rightarrow (1)$$

$$z = e^u$$

$$\frac{\partial z}{\partial x} = e^u \frac{\partial u}{\partial x}$$

$$\frac{\partial z}{\partial y} = e^u \frac{\partial u}{\partial y}$$

$$(1) \Rightarrow x \cdot e^u \frac{\partial u}{\partial x} + y\, e^u \frac{\partial u}{\partial y} = e^u$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 1$$

Hence proved

Q) If $u = \cot^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$ S.T $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{4} \sin 2u = 0$

$$\cot u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

let $z = \cot u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$

clearly z is a homogeneous function of degree $n = \frac{1}{2}$

$\therefore$ By Euler's theorem $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{1}{2} z \rightarrow (1)$

$$z = \cot u$$

$$\frac{\partial z}{\partial x} = -\operatorname{cosec}^2 u \frac{\partial u}{\partial x}$$

$$\frac{\partial z}{\partial y} = -\operatorname{cosec}^2 u \frac{\partial u}{\partial x}$$

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$$(1) \Rightarrow -\operatorname{cosec}^2 u \frac{\partial u}{\partial x} - y \operatorname{cosec}^2 u \frac{\partial u}{\partial y} = \frac{1}{2}\cot u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \frac{\cot u}{\operatorname{cosec}^2 u}$$

$$= -\frac{1}{2} \frac{\cos u}{\sin u} \sin^2 u$$

$$= -\frac{1}{2} \cos u \sin u = -\frac{1}{2} \frac{\sin 2u}{2} = -\frac{1}{4} \sin 2u$$

$$x \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{1}{4} \sin 2u = 0$$

Total derivative: If $u = f(x,y)$ and $x = \phi(t)$; $y = \psi(t)$ then the total derivative of u is given by

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \quad \text{(or)} \quad du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} \cdot dy$$

chain rule:

$u = f(x,y)$; $x = \phi(r,s)$; $y = \psi(r,s)$ then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial s}$$

1) If $u = f(2x-3y, 3y-4z, 4z-2x)$ P.T $\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$

let $r = 2x-3y$, $s = 3y-4z$, $t = 4z-2x$

$$\left\{\frac{\partial r}{\partial x} = 2 ; \frac{\partial r}{\partial y} = -3 ; \frac{\partial r}{\partial z} = 0\right\} \left\{\frac{\partial s}{\partial x} = 0 ; \frac{\partial s}{\partial y} = 3 ; \frac{\partial s}{\partial z} = -4\right\}$$

$$\left\{\frac{\partial t}{\partial x} = -2 ; \frac{\partial t}{\partial y} = 0 ; \frac{\partial t}{\partial z} = 4\right\}$$

$u = f(r,s,t)$ by chain rule

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial x} = 2 \cdot \frac{\partial u}{\partial r} + 0 \cdot \frac{\partial u}{\partial s} + -2 \frac{\partial u}{\partial t}$$

$$\frac{1}{2} \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \rightarrow (1)$$

$$\frac{\partial U}{\partial y} = \frac{\partial U}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial U}{\partial s}\cdot\frac{\partial s}{\partial y} + \frac{\partial U}{\partial t}\cdot\frac{\partial t}{\partial y}$$

since

$$= -3\frac{\partial U}{\partial r} + 3\cdot\frac{\partial U}{\partial s} + 0\cdot\frac{\partial U}{\partial t}$$

$$\frac{1}{3}\frac{\partial U}{\partial y} = \frac{\partial U}{\partial s} - \frac{\partial U}{\partial r} \rightarrow ②$$

$$\frac{\partial U}{\partial z} = \frac{\partial U}{\partial r}\cdot\frac{\partial r}{\partial z} + \frac{\partial U}{\partial s}\cdot\frac{\partial s}{\partial z} + \frac{\partial U}{\partial t}\frac{\partial t}{\partial z}$$

$$= 4\frac{\partial U}{\partial r} + (-4)\frac{\partial U}{\partial s} + 4\frac{\partial U}{\partial t}$$

$$\frac{1}{4}\frac{\partial U}{\partial z} = \frac{\partial U}{\partial t} - \frac{\partial U}{\partial s} \rightarrow ③$$

1+2+3

$$\frac{1}{2}\frac{\partial U}{\partial x} + \frac{1}{3}\frac{\partial U}{\partial y} + \frac{1}{4}\frac{\partial U}{\partial z} = \frac{\partial U}{\partial r} - \frac{\partial U}{\partial t} + \frac{\partial U}{\partial s} - \frac{\partial U}{\partial r} + \frac{\partial U}{\partial t} - \frac{\partial U}{\partial s}$$

$$\frac{1}{2}\frac{\partial U}{\partial x} + \frac{1}{3}\frac{\partial U}{\partial y} + \frac{1}{4}\frac{\partial U}{\partial z} = 0$$ Hence prooved.

3) $u = f(r)$; $x = r\cos\theta$; $y = r\sin\theta$ P.T $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$

Sol) Given $u = f(r)$

By the total derivative

$$\frac{\partial U}{\partial x} = f'(r)\cdot\frac{\partial r}{\partial x}$$

$$\frac{\partial^2 U}{\partial x^2} = \left(f''(r)\frac{\partial r}{\partial x}\right)\frac{\partial r}{\partial x} + f'(r)\frac{\partial^2 r}{\partial x^2}$$

$$\frac{\partial^2 u}{\partial x^2} = f''(r)\left(\frac{\partial r}{\partial x}\right)^2 + f'(r)\frac{\partial^2 r}{\partial x^2}$$

similarly

$$\frac{\partial^2 U}{\partial y^2} = f''(r)\left(\frac{\partial r}{\partial y}\right)^2 + f'(r)\left(\frac{\partial^2 r}{\partial y^2}\right)$$

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = f''(r)\left(\frac{\partial r}{\partial x}\right)^2 + f'(r)\left(\frac{\partial^2 r}{\partial x^2}\right) + f''(r)\left(\frac{\partial r}{\partial y}\right)^2 + f'(x)\frac{\partial^2 r}{\partial y^2}$$

$$= f''(r)\left[\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2\right] + f'(r)\left[\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2}\right] \rightarrow ①$$

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since $x = r\cos\theta,\ y = r\sin\theta$

$$r^2 = x^2 + y^2$$

$$2r \cdot \frac{\partial r}{\partial x} = 2x$$

$$\boxed{\frac{\partial r}{\partial x} = \frac{x}{r}} \quad ||^{ly} \quad \boxed{\frac{\partial r}{\partial y} = \frac{y}{r}}$$

$$\frac{\partial^2 r}{\partial x^2} = \frac{r \cdot 1 - x\frac{\partial r}{\partial x}}{r} = \frac{1}{r^2}\left[r - \frac{x^2}{r}\right]$$

$$\boxed{\frac{\partial^2 r}{\partial x^2} = \frac{r^2 - x^2}{r^3}} \quad \& \quad \boxed{\frac{\partial^2 r}{\partial y^2} = \frac{r^2 - y^2}{r^3}}$$

$$① \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r)\left[\frac{x^2}{r^2} + \frac{y^2}{r^2}\right] + f'(r)\left[\frac{r^2 - x^2}{r^3} + \frac{r^2 - y^2}{r^3}\right]$$

$$= f''(r)\left[\frac{x^2+y^2}{r^2}\right] + f'(r)\left[\frac{2r^2 - (x^2+y^2)}{r^3}\right] = f''(r) + f'(r)\frac{1}{r}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$$

Hence proved

Jacobian:

let $u = f(x,y),\ v = g(x,y)$ be the two functions of x, y then the jacobian of u, v w.r.t $x \& y$ is given by $J\left(\frac{u,v}{x,y}\right)$ or $\frac{\partial(u,v)}{\partial(x,y)}$

(or) $$\partial\left(\frac{u,v}{x,y}\right) = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

Properties:

i) P.T $J \cdot J' = 1$ (or) $u = f(x,y)$ & $v = g(x,y)$ then

$$\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = 1$$

Proof: let $u = f(x,y)$ and $v = g(x,y)$ then

$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

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By the transformation let $x = \phi(u,v)$; $y = \psi(u,v)$ then

$$J' = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

① × ②

$$J \cdot J' = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial v} \\ \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial v} \end{vmatrix}$$

By the chain rule

$$= \begin{vmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial v} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$J \cdot J' = 1$

Q) If $u = f(x,y)$; $v = g(x,y)$ & $x = \phi(r,s)$; $y = \psi(r,s)$ then

P.T $\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(r,s)} = \frac{\partial(u,v)}{\partial(r,s)}$

Sol) let $u = f(x,y)$; $v = g(x,y)$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \rightarrow ①$$

let $x = \phi(r,s)$; $y = \psi(r,s)$

$$J' = \frac{\partial(x,y)}{\partial(r,s)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} \end{vmatrix} \rightarrow ②$$

① × ②

$$\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(r,s)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial r} & \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial s} \\ \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial r} & \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial s} \end{vmatrix}$$

By chain rule

$$= \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} = \frac{\partial(u,v)}{\partial(r,s)}$$ Hence proved.

Functionally dependent:

If $u = f(x,y)$ and $v = g(x,y)$ then u, v are said to be functionally dependent if jacobian vanishes i.e $\frac{\partial(u,v)}{\partial(x,y)} = 0$

Q) If $u = \frac{y}{x}$, $v = xy$ find $\frac{\partial(u,v)}{\partial(x,y)}$

so) $$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

given $u = \frac{y}{x}$; $v = xy$

$\frac{\partial u}{\partial x} = \frac{-y}{x^2}$; $\frac{\partial v}{\partial x} = y$

$\frac{\partial u}{\partial y} = \frac{1}{x}$; $\frac{\partial v}{\partial y} = x$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{-y}{x^2} & \frac{1}{x} \\ y & x \end{vmatrix} = \frac{-y}{x} - \frac{y}{x} = \frac{-2y}{x}$$

Q) $x = uv$, $y = \frac{u+v}{u-v}$ find $\frac{\partial(u,v)}{\partial(x,y)}$

A) Given $x = uv$; $y = \frac{u+v}{u-v}$

$\frac{\partial x}{\partial u} = v$; $\frac{\partial x}{\partial v} = u$

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$$\frac{\partial y}{\partial u} = \frac{(u-v)\frac{\partial}{\partial u}(u+v) - (u+v)\frac{\partial}{\partial u}(u-v)}{(u-v)^2} = \frac{u-v-u+v}{(u-v)^2} =$$

$$\frac{\partial y}{\partial v} = \frac{(u-v)\frac{\partial}{\partial v}(u+v) - (u+v)\frac{\partial}{\partial v}(u-v)}{(u-v)^2} = \frac{(u-v)(1)-(u+v)(-1)}{(u-v)^2} = \frac{2u}{(u-v)^2}$$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ \frac{-2v}{(u-v)^2} & \frac{2u}{(u-v)^2} \end{vmatrix}$$

$$\frac{\partial(x,y)}{\partial(u,v)} = \frac{2uv}{(u-v)^2} + \frac{2uv}{(u-v)^2} = \frac{4uv}{(u-v)^2}$$

By jacobion 1st property

$$\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{(u-v)^2}{4uv}$$

Q) If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$ find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$

Sol) $u = \frac{yz}{x}$ | $v = \frac{zx}{y}$ | $w = \frac{xy}{z}$

$\frac{\partial u}{\partial x} = -\frac{yz}{x^2}$ | $\frac{\partial v}{\partial x} = \frac{z}{y}$ | $\frac{\partial w}{\partial x} = \frac{y}{z}$

$\frac{\partial u}{\partial y} = \frac{z}{x}$ | $\frac{\partial v}{\partial y} = -\frac{zx}{y^2}$ | $\frac{\partial w}{\partial y} = \frac{x}{z}$

$\frac{\partial u}{\partial z} = \frac{y}{x}$ | $\frac{\partial v}{\partial z} = \frac{x}{y}$ | $\frac{\partial w}{\partial z} = -\frac{xy}{z^2}$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & -\frac{zx}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & -\frac{xy}{z^2} \end{vmatrix}$$

$$= -\frac{yz}{x^2}\left\{\frac{x^2yz}{z^2y^2} - \frac{x^2}{yz}\right\} - \frac{z}{x}\left\{\frac{-xyz}{yz^2} - \frac{xy}{zy}\right\} + \frac{y}{x}\left\{\frac{zx}{yz} + \frac{xyz}{y^2z}\right\}$$

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$$= \frac{-yz}{x^2}\left(\frac{x^2-x^2}{yz}\right) - \frac{z}{x}\left(\frac{-2xy}{zy}\right) + \frac{y}{x}\left(\frac{2zx}{zy}\right)$$

$$= 2+2 = 4$$

If $x+y+z = u$, $y+z = uv$, $z = uvw$ find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$ also find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$

$z = uvw$; $y+z = uv$; $x+y+z = u$

$y + uvw = uv$; $x = u - vu$

$\boxed{y = uv - uvw}$

$x = u - vu \Rightarrow \frac{\partial x}{\partial u} = 1-v$; $\frac{\partial x}{\partial v} = -u$; $\frac{\partial x}{\partial w} = 0$

$y = uv - uvw \Rightarrow \frac{\partial y}{\partial u} = v - vw$; $\frac{\partial y}{\partial v} = u - uw$; $\frac{\partial z}{\partial v} = uw$

$z = uvw \Rightarrow \frac{\partial z}{\partial u} = vw$; $\frac{\partial z}{\partial v} = uw$; $\frac{\partial z}{\partial w} = uv$

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} 1-v & -u & 0 \\ v-vw & u-uw & -uv \\ vw & uw & uv \end{vmatrix} = (1-v)\{(u-uw)(uv) + (uv)(uw)\} + u((v-vw)(uv) + (vw)(uv)) + 0$$

$$= (1-v)\{(u^2v - u^2vw) + u^2vw\} + u\{uv^2 - uv^2w + uv^2w\}$$

$$= (1-v)(u^2v) + u^2v^2 = u^2v - u^2v^2 + u^2v^2 = u^2v$$

By jacobian 1st property

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} \cdot \frac{\partial(u,v,w)}{\partial(x,y,z)} = 1$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \frac{1}{u^2v}$$

If $x = r\cos\theta$; $y = r\sin\theta$ Prove that $J.J' = 1$

Given $x = r\cos\theta$; $y = r\sin\theta$

$\frac{\partial x}{\partial r} = \cos\theta$ | $\frac{\partial y}{\partial r} = \sin\theta$

$\frac{\partial x}{\partial \theta} = -r\sin\theta$ | $\frac{\partial y}{\partial \theta} = r\cos\theta$

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r(\sin^2\theta + \cos^2\theta)$$

$$= r$$

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To find J'

since $x = r\cos\theta$, $y = r\sin\theta$

$x^2+y^2=r^2$

$2x = 2r\frac{\partial r}{\partial x}$

$\frac{\partial r}{\partial x} = \frac{x}{r}$

$\frac{y}{x} = \frac{r\sin\theta}{r\cos\theta}$

$\frac{y}{x} = \tan\theta$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$$\frac{\partial\theta}{\partial x} = \frac{1}{1+\left(\frac{y}{x}\right)^2}\left(\frac{-y}{x^2}\right) = \frac{x^2}{x^2+y^2}\left(\frac{-y}{x^2}\right) = \frac{-y}{x^2+y^2}$$

$$\frac{\partial\theta}{\partial y} = \frac{1}{1+\left(\frac{y}{x}\right)^2}\left(\frac{1}{x}\right) = \frac{x^2}{x^2+y^2}\cdot\frac{1}{x} = \frac{x}{x^2+y^2}$$

$$\frac{\partial(r,\theta)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial\theta}{\partial x} & \frac{\partial\theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{r} & \frac{y}{r} \\ \frac{-y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{vmatrix} = \frac{x^2}{r(x^2+y^2)} + \frac{y^2}{r(x^2+y^2)}$$

$$= \frac{x^2+y^2}{r(x^2+y^2)} = \frac{1}{r}$$

$$J \cdot J' = r\cdot\frac{1}{r} = 1$$

Hence proved

If $u = \frac{x+y}{1-xy}$; $v = \tan^{-1}x + \tan^{-1}y$ find $\frac{\partial(u,v)}{\partial(x,y)}$

$$\frac{\partial u}{\partial x} = \frac{(1-xy)\frac{\partial}{\partial x}(x+y) - (x+y)\frac{\partial}{\partial x}(1-xy)}{(1-xy)^2}$$

$$= \frac{(1-xy)(1) - (x+y)(-y)}{(1-xy)^2} = \frac{1-xy+xy+y^2}{(1-xy)^2} = \frac{1+y^2}{(1-xy)^2}$$

$$\frac{\partial u}{\partial y} = \frac{(1-xy)\frac{\partial}{\partial y}(x+y) - (x+y)\frac{\partial}{\partial y}(1-xy)}{(1-xy)^2}$$

$$= \frac{(1-xy)(1) - (x+y)(-x)}{(1-xy)^2} = \frac{1-\cancel{xy}+x^2+\cancel{xy}}{(1-xy)^2} = \frac{1+x^2}{(1-xy)^2}$$

$\frac{\partial v}{\partial x} = \tan^{-1}x + \tan^{-1}y$

$= \frac{1}{\ldots}$

$\frac{\partial v}{\partial y} = \frac{1}{1+y^2}$

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$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{1+y^2}{(1-xy)^2} & \frac{1+x^2}{(1-xy)^2} \\ \frac{1}{1+x^2} & \frac{1}{1+y^2} \end{vmatrix}$$

$$= \frac{\cancel{1+y^2}}{(1-xy)^2(\cancel{1+y^2})} - \frac{\cancel{1+x^2}}{(1-xy)^2(\cancel{1+x^2})}$$

$$= \frac{1}{(1-xy)^2} - \frac{1}{(1-xy)^2} = 0$$

Q) $u = x^2 - y^2$, $v = 2xy$ and $x = r\cos\theta$, $y = r\sin\theta$ find $\frac{\partial(u,v)}{\partial(r,\theta)}$

Sol) By the jacobian second property

$$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(r,\theta)}$$

To find $\frac{\partial(u,v)}{\partial(x,y)}$

$\frac{\partial u}{\partial x} = 2x$ | $\frac{\partial u}{\partial y} = -2y$

$\frac{\partial v}{\partial x} = 2y$ | $\frac{\partial v}{\partial y} = 2x$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 2x & 2y \\ -2y & 2x \end{vmatrix}$$

$$= 4x^2 + 4y^2 = 4(x^2+y^2)$$

find $\frac{\partial(x,y)}{\partial(r,\theta)}$

$x = r\cos\theta$ | $y = r\sin\theta$

$\frac{\partial x}{\partial r} = \cos\theta$ | $\frac{\partial y}{\partial r} = \sin\theta$

$\frac{\partial x}{\partial \theta} = -r\sin\theta$ | $\frac{\partial y}{\partial \theta} = r\cos\theta$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta$$

$$= r$$

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$$\frac{\partial(U,V)}{\partial(r,\theta)} = r\left[4\cdot(x^2+y^2)\right]$$
$$= 4r(x^2+y^2) = 4r(r^2) = 4r^3$$

Q) $x = e^r\sec\theta$, $y = e^r\tan\theta$ Prove that $J \cdot J' = 1$

sol) To find J

$x = e^r\sec\theta$; $y = e^r\tan\theta$

$\frac{\partial x}{\partial r} = e^r\sec\theta$ $\qquad$ $\frac{\partial y}{\partial r} = e^r\tan\theta$

$\frac{\partial x}{\partial \theta} = e^r\sec\theta\tan\theta$ $\qquad$ $\frac{\partial y}{\partial \theta} = e^r\sec^2\theta$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} e^r\sec\theta & e^r\sec\theta\tan\theta \\ e^r\tan\theta & e^r\sec^2\theta \end{vmatrix} = e^{2r}\sec^3\theta - e^{2r}\sec\theta\tan^2\theta$$
$$= e^{2r}\sec\theta\left[\sec^2\theta - \tan^2\theta\right]$$
$$= e^{2r}\sec\theta$$

To find $J' = \frac{\partial(r,\theta)}{\partial(x,y)}$

Given $x = e^r\sec\theta$; $y = e^r\tan\theta$

squaring and subtracting

$$x^2 - y^2 = e^{2r}\left[\sec^2\theta - \tan^2\theta\right]$$
$$x^2 - y^2 = e^{2r}$$

$$\frac{y}{x} = \frac{\tan\theta}{\sec\theta} = \sin\theta$$

$$\theta = \sin^{-1}\left(\frac{y}{x}\right)$$

$e^{2r} = x^2 - y^2$

$2e^{2r}\frac{\partial r}{\partial x} = 2x$

$\frac{\partial r}{\partial x} = \frac{x}{e^{2r}} = xe^{-2r}$

$\theta = \sin^{-1}\left(\frac{y}{x}\right)$

$\frac{\partial \theta}{\partial x} = \frac{1}{\sqrt{1 - \frac{y^2}{x^2}}}\left(-\frac{y}{x^2}\right)$

$= \frac{-y}{x\sqrt{x^2-y^2}}$

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$2e^{2r}\frac{\partial r}{\partial y}=-2y$ | $\frac{\partial \theta}{\partial y}=\frac{1}{\sqrt{1-\frac{y^2}{x^2}}}\left(\frac{1}{x}\right)=\frac{1}{\sqrt{x^2-y^2}}$

$\frac{\partial r}{\partial y}=-y\cdot e^{-2r}$

$$\frac{\partial(r,\theta)}{\partial(x,y)}=\frac{xe^{-2r}}{\sqrt{x^2-y^2}}-\frac{y^2e^{-2r}}{x\sqrt{x^2-y^2}}$$

$$=\frac{x^2e^{-2r}-y^2e^{-2r}}{x\sqrt{x^2-y^2}}=\frac{e^{-2r}(x^2-y^2)}{x\sqrt{x^2+y^2}}=\frac{e^{-2r}\cdot e^{2r}}{x\cdot e^{r}}$$

$$=\frac{e^{-r}}{x}=\frac{e^{-r}}{e^{r}\sec\theta}$$

$$J\cdot J'=e^{2r}\sec\theta\cdot\frac{e^{-r}}{e^{r}\sec\theta}=e^{2r}\sec\theta\cdot\frac{1}{e^{2r}\sec\theta}=1$$

2) If $u=\frac{x+y}{1-xy}$; $v=\tan^{-1}x+\tan^{-1}y$ prove that u & v are functionally dependent and find the relation between them.

sol) $$\frac{\partial u}{\partial x}=\frac{(1-xy)\frac{\partial}{\partial x}(x+y)-(x+y)\frac{\partial}{\partial x}(1-xy)}{(1-xy)^2}$$

$$=\frac{(1-xy)-(x+y)(-y)}{(1-xy)^2}=\frac{1-xy+xy+y^2}{(1-xy)^2}=\frac{1+y^2}{(1-xy)^2}$$

$$\frac{\partial u}{\partial y}=\frac{(1-xy)\frac{\partial}{\partial y}(x+y)-(x+y)\frac{\partial}{\partial y}(1-xy)}{(1-xy)^2}$$

$$=\frac{(1-xy)-(x+y)(-x)}{(1-xy)^2}=\frac{1-xy+x^2+xy}{(1-xy)^2}=\frac{1+x^2}{(1-xy)^2}$$

$$\frac{\partial v}{\partial x}=\tan^{-1}x+\tan^{-1}y=\frac{1}{1+x^2}$$

$$\frac{\partial v}{\partial y}=\frac{1}{1+y^2}$$

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$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{1+y^2}{(1-xy)^2} & \frac{1+x^2}{(1-xy)^2} \\ \frac{1}{1+x^2} & \frac{1}{1+y^2} \end{vmatrix} = \frac{1+y^2}{(1+y^2)(1-xy)^2} - \frac{1+x^2}{(1+x^2)(1-xy)^2}$$

$$= \frac{1}{(1-xy)^2} - \frac{1}{(1-xy)^2} = 0$$

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$v = \tan^{-1}u$$

Maximum and minimum for function of 2 variables:

let $f(x,y)$ be a function of 2 variables then $f(a,b)$ is said to be maximum or minimum value if $f(a,b) > f(a+h, b+k)$ or $f(a,b) < f(a+h,b+k)$ respectively

Procedure:

step 1: Find $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ $\quad l = r = \frac{\partial^2 f}{\partial x^2}$, $m = s = \frac{\partial^2 f}{\partial xy}$

$$n = t = \frac{\partial^2 f}{\partial y^2}$$

step 2: consider $\frac{\partial f}{\partial x} = 0$; $\frac{\partial f}{\partial y} = 0$; on solving these two equations we get set of points (a_1,b_1) (a_2,b_2) ---- known as stationary points

step 3: At all the stationary points find l, m, n

step 4: If at some stationary point (a_i, b_i)

a) $ln - m^2 > 0$; $l < 0$ then (a_i, b_i) is maximum point and $f(a_i,b_i)$ is the maximum value

b) If $ln - m^2 > 0$; $l > 0$ then (a_i,b_i) is minimum point and $f(a_i,b_i)$ is the minimum value

c) If $ln - m^2 < 0$; then (a_i,b_i) is neither maximum nor minimum known <u>saddle point</u>.

d) If $ln - m^2 = 0$ then no conclusion

Extremum point: If a stationary point is either maximum or minimum then it is known as extremum point and the value is known as extremum value

Find the maxima and minima of $f = x^3 + y^3 - 3axy$

$\frac{\partial f}{\partial x} = 3x^2 + 0 - 3ay = 3x^2 - 3ay$; $\frac{\partial f}{\partial y} = 0 + 3y^2 - 3ax = 3y^2 - 3ax$

$l = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}(3x^2 - 3ay) = 6x$

$m = \frac{\partial^2 f}{\partial x \cdot \partial y} = \frac{\partial}{\partial x}(3y^2 - 3ax) - 3a$

$n = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}(3y^2 - 3ax) = 6y$

To find stationary points:

$\frac{\partial f}{\partial x} = 0$	$\frac{\partial f}{\partial y} = 0$
$3x^2 - 3ay = 0$	$3y^2 - 3ax = 0$
$x^2 - ay = 0$	$y^2 - ax = 0$
$x^2 = ay$	$\frac{x^2}{a^2} - ax = 0$
	$x^4 - a^3 x = 0$
	$x = 0 ; x = a$

for $x = 0$; $y = 0$

for $x = a$; $y^2 - a^2 = 0$

$y = a$

$\therefore$ stationary points are $(0,0)$ $(0,a)$

at $(0,0)$

$l = 0, m = 3a; n = 0$

$ln - m^2 = 0 - (-3a)^2 = -9a^2 < 0$

$(0,0)$ is a saddle point.

at (a,a)

$l = 6a; m = -3a; n = 6a$

$ln - m^2 = 36a^2 - 9a^2 = 27a^2 > 0$ & $l > 0$

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(a,a) is minimum point & min value

$f(a,a) = a^3+a^3-3(a)(a)(a) = a^3+a^3-3a^3 = 2a^3-3a^3 = -a^3$

Q) $f = x^3+3xy^2-3x^2-3y^2+4$.

sol) $\frac{\partial f}{\partial x} = 3x^2+3y^2-6x$

$\frac{\partial f}{\partial y} = 0+6xy-6y = 6xy-6y$

$l = \frac{\partial^2 f}{\partial x^2} = 6x-6$

$m = \frac{\partial^2 f}{\partial x \cdot \partial y} = \frac{\partial}{\partial x}(6xy-6y) = 6y$

$n = \frac{\partial^2 f}{\partial y^2} = 6x-6$

To find stationary point

$\frac{\partial f}{\partial x} = 0$

$3x^2+3y^2-6x = 0$

$x^2+y^2-2x = 0 \rightarrow ①$

$\frac{\partial f}{\partial y} = 6xy-6y = 0$

$6y(x-1) = 0$

$y = 0;\ x = 1$

when $y = 0 \Rightarrow x^2-2x = 0$

$x(x-2) = 0$

$(0,0)\ (2,0)$

$x = 1;\quad 1+y^2 = 2$

$y^2-1 = 0$

$y = \pm 1$

$(1,1)\ (1,-1)$

$(0,0)\ (2,0)\ (1,1)\ (1,-1)$ are stationary points

at $(0,0)$

$l = -6,\ m = 0,\ n = -6$

$ln-m^2 = 36 > 0 \quad l < 0$

$(0,0)$ max point; value of $f(0,0) = 4$

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(3)

$y < \pi$

$)\cos x] = \sin y \sin(2x+y)$

$\sin x [\sin(x+2y)] = 2\sin y \cos(2x+y)$

$) = \sin x \sin(x+2y)$

$= \sin(2x+2y)$

$= 2\sin x \cos(x+2y)$

$\frac{\partial f}{\partial x} = 0$

$= \pi$ — (5)

we cannot take the

so we cannot take tha

$2x+y=0$ & $x+2y=0$ we get $x=0, y=0$

points also

solving ① & ②

$$2x+y=\pi$$
$$2x+4y=2\pi$$
$$-3y=-\pi$$
$$y=\frac{\pi}{3}$$
$$2x+y=\pi$$
$$2x=\frac{3\pi-\pi}{3}$$
$$\boxed{x=\frac{\pi}{3}}$$

$\left(\frac{\pi}{3},\frac{\pi}{3}\right)$ stationary points

at $\frac{\pi}{3},\frac{\pi}{3}$

$l=-\sqrt{3}$ $\quad m=-\frac{\sqrt{3}}{2}$ $\quad n=-\sqrt{3}$

$$ln-m^2=-\sqrt{3}\times-\sqrt{3}-\left(\frac{\sqrt{3}}{2}\right)^2=3-\frac{3}{4}=\frac{12-3}{4}=\frac{9}{4}>0$$

maximum point

$$\sin\left(\frac{\pi}{3}\right)\sin\left(\frac{\pi}{3}\right)\sin\left(\frac{2\pi}{3}\right)$$

maximum value $=0$

Lagrange's function:

let $f(x,y,z)$ be the function of 2 variables, subject to $\phi(x,y,z)=0$ Then extremum value of n can be obtained by using lagrange's function.

STEP 1:

construct the

$$F(x,y,z)=f(x,y,z)+\lambda\phi(x,y,z)$$

where λ is known as lagrange's multiplier

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Take the STEP-2:

Find a) $\frac{\partial f}{\partial x}=0 \Rightarrow \frac{\partial f}{\partial x}+\lambda\frac{\partial \phi}{\partial x}=0$

b) $\frac{\partial f}{\partial y}=0 \Rightarrow \frac{\partial f}{\partial y}+\lambda\frac{\partial \phi}{\partial y}=0$

c) $\frac{\partial f}{\partial z}=0 \Rightarrow \frac{\partial f}{\partial z}+\lambda\frac{\partial \phi}{\partial z}=0$

solving these three equation's we get the relation between (x,y,z) substituting in ϕ gives the values of (x,y,z)

Q) Find the minimum value of $x^2+y^2+z^2$; $\phi = x+y+z-3a=0$

Sol) Given $x+y+z=3a$

let $f=x^2+y^2+z^2$ & $\phi = x+y+z-3a=0$

lagrange's function $F=f+\lambda\phi$

$$F=(x^2+y^2+z^2)+\lambda(x+y+z-3a)$$

$\frac{\partial F}{\partial x}=0 \Rightarrow 2x+\lambda=0 \Rightarrow \lambda=-2x \rightarrow (1)$

$\frac{\partial F}{\partial y}=0 \Rightarrow 2y+\lambda=0 \Rightarrow \lambda=-2y \rightarrow (2)$

$\frac{\partial F}{\partial z}=0 \Rightarrow 2z+\lambda=0 \Rightarrow \lambda=-2z \rightarrow (3)$

from (1) (2) & (3)

$$-2x=-2y=-2z$$
$$x=y=z$$

$x+y+z=3a \Rightarrow 3x=3a \Rightarrow x=a, y=a, z=a$

$\therefore (a,a,a)$ is the minimum point & min value is

$$f=a^2+a^2+a^2=3a^2$$

Find the max & min values of $x+y+z$ subject to $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$

$f=x,y,z$; $\phi=\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$

$F=f+\lambda\phi$

$$F=(x+y+z)+\lambda\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$$

$\frac{\partial F}{\partial x} = 0 \Rightarrow 1 - \frac{\lambda}{x^2} = 0 \Rightarrow \lambda = x^2 \Rightarrow x = \pm\sqrt{\lambda}$

$\frac{\partial F}{\partial y} = 0 \Rightarrow 1 - \frac{\lambda}{y^2} = 0 \Rightarrow \lambda = y^2 \Rightarrow y = \pm\sqrt{\lambda}$

$\frac{\partial F}{\partial z} = 0 \Rightarrow 1 - \frac{\lambda}{z^2} = 0 \Rightarrow \lambda = z^2 \Rightarrow z = \pm\sqrt{\lambda}$

$(x,y,z) = (-\sqrt{\lambda}, -\sqrt{\lambda}, -\sqrt{\lambda})$

$-\frac{1}{\sqrt{\lambda}} - \frac{1}{\sqrt{\lambda}} - \frac{1}{\sqrt{\lambda}} = 1$

$\frac{-3}{\sqrt{\lambda}} = 1$

$-3 = \sqrt{\lambda}$

$(x,y,z) = (3,3,3)$ max point

-3 is min point

$(x,y,z) = (\sqrt{\lambda}, \sqrt{\lambda}, \sqrt{\lambda})$

$= \frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{\lambda}}$

$= \frac{3}{\sqrt{\lambda}} = 1$

$\sqrt{\lambda} = 3$

$(x,y,z) = (-3,-3,-3)$

Q) Find the volume of the Rectangules parallelopiped that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

A) let $2x, 2y, 2z$ be the dimensions of the parallelopipe then

volume $v = 8xyz$

let $F = 8xyz$ & $\phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$

$F = 8xyz + \lambda\left[\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right]$

$\frac{\partial F}{\partial x} = 0 \Rightarrow 8yz + \frac{2x}{a^2}\lambda = 0 \Rightarrow \lambda = -\frac{4yza^2}{x} \rightarrow (1)$

$\frac{\partial F}{\partial y} = 0 \Rightarrow 8xz + \frac{2y}{b^2}\lambda = 0 \Rightarrow \lambda = \frac{-4xzb^2}{y} \rightarrow (2)$

$\frac{\partial F}{\partial z} = 0 \Rightarrow 8xy + \frac{2z}{c^2}\lambda = 0 \Rightarrow \lambda = \frac{-4xyc^2}{z} \rightarrow (3)$

(1) & (2)

$\frac{-4yza^2}{x} = \frac{-4xzb^2}{y}$

$\frac{y^2}{b^2} = \frac{x^2}{a^2}$

② + ③ $\frac{-4xzb^2}{y} = \frac{-4xy}{z}c^2$

$$\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2}$$

sob in 'ϕ'

$\frac{3x^2}{a^2} = 1 \Rightarrow x = \frac{a}{\sqrt{3}}$; $||^{ly}$ $y = \frac{b}{\sqrt{3}}$; $z = \frac{c}{\sqrt{3}}$

volume $(v) = \frac{8abc}{3\sqrt{3}}$

Q) Find the 3 positive no's the product is max & sum is 100

Sol) let x, y, z be 3 +ve no's

$F = xyz$ is max & $\phi = x+y+z = 100$

$F = xyz + \lambda(x+y+z-100)$

$\frac{\partial F}{\partial x} = 0 \Rightarrow yz + \lambda = 0 \Rightarrow \lambda = -yz \rightarrow ①$

$\frac{\partial F}{\partial y} = 0 \Rightarrow xz + \lambda = 0 \Rightarrow \lambda = -xz \rightarrow ②$

$\frac{\partial F}{\partial z} = 0 \Rightarrow xy + \lambda = 0 \Rightarrow \lambda = -xy \rightarrow ③$

① & ② $x = y$; ② & ③ $z = y$

$x+y+z = 100$

$3y = 100 \Rightarrow y = \frac{100}{3}$

Q) Find the point on the sphere $x^2+y^2+z^2=4$ which is closest and farthest from the point $(3,1,-1)$

Sol) let $P(x,y,z)$ be the point on the sphere

then $AP^2 = (x-3)^2 + (y-1)^2 + (z+1)^2$

let $f = AP^2 = (x-3)^2 + (y-1)^2 + (z+1)^2$ & $\phi = x^2+y^2+z^2-4 = 0$

$$F = (x-3)^2 + (y-1)^2 + (z+1)^2 + \lambda(x^2+y^2+z^2-4)$$

$$\frac{\partial F}{\partial x} = 2(x-3) + 2x\lambda = 0$$

$$\lambda = \frac{-(x-3)}{x} \Rightarrow \lambda = \frac{3}{x} - 1 \rightarrow (1)$$

$$\frac{\partial F}{\partial y} = 2(y-1) + 2y\lambda \Rightarrow \lambda = \frac{1}{y} - 1 \rightarrow (2)$$

$$\frac{\partial F}{\partial z} = 2(z+1) + 2z\lambda \Rightarrow \lambda = -\frac{1}{z} - 1 \rightarrow (3)$$

(1) & (2) $\quad \frac{3}{x} - 1 = \frac{1}{y} - 1$

$$x = 3y$$

2 & 3 $\quad \frac{1}{y} - 1 = -\frac{1}{z} - 1$

$$z = -y$$

$$x^2 + y^2 + z^2 = 4$$

$$9y^2 + y^2 + y^2 = 4$$

$$y^2 = \frac{4}{11} \Rightarrow y = \pm \frac{2}{\sqrt{11}}$$

$$x = \pm \frac{6}{\sqrt{11}}$$

$$z = \frac{\pm 2}{\sqrt{11}}$$

near pt $\left(-\frac{6}{\sqrt{11}}, \frac{-2}{\sqrt{11}}, \frac{2}{\sqrt{11}}\right)$

far point $\left(\frac{6}{\sqrt{11}}, \frac{2}{\sqrt{11}}, \frac{-2}{\sqrt{11}}\right)$